Range Ambiguity Detection and Suppression in Spaceborne SAR Image via Image Post-Processing

Jieshuang Li, Yanyang Liu, Mingliang Tao, *Senior Member, IEEE*, Tao Li, Junli Chen, and Ling Wang, *Member, IEEE*

***Abstract*— Range ambiguity is a common issue for spaceborne synthetic aperture radar (SAR) systems, leading to image quality degradation and subsequent interpretation accuracy. Existing range ambiguity suppression methods mainly rely on raw echo domain processing with specific requirements on prior knowledge. However, most users can only obtain single-look-complex (SLC) products instead of raw echo products. Raw echo obtained from SLC through an inverse focusing process will waste additional time and space resources. This article proposes a novel scheme to detect and suppress range ambiguity in SLC products to deal with this deficiency. The proposed method designs a detector to extract and suppress the focused range ambiguity based on the characteristic difference between the desired signal and range ambiguity in the fractional transform domain of the SLC image. Experimental results on real measured L-band spaceborne interferometry SAR system verify the detection performance of the proposed method, which is beneficial for subsequent land monitoring applications.**

***Index Terms*— Fractional Fourier transform (FRFT), range ambiguity, single-look-complex (SLC), synthetic aperture radar (SAR).**

I. Introduction

A. Background and Motivation

SYNTHETIC aperture radar (SAR) is an active microwave remote sensing device. With its powerful penetration ability, it can achieve high-precision, all-time, all-weather surface deformation monitoring, and terrain mapping of targeted areas [1]. However, ground systems occasionally receive images of the spaceborne SAR marred by ghost artifacts. Among these artifacts, range ambiguity emerges as a particularly severe issue [2].

In traditional SAR system, it is difficult to achieve high resolution and wide swath at the same time due to the constraint of the minimum antenna area [3]. Under the low pulse repetition interval (PRI), it will meet the wide swath and avoid azimuth ambiguity, but give rise to range ambiguity. Besides, the echo of the previous PRI received by the antenna sidelobe can also cause range ambiguity. The range ambiguity will lead to image quality distortion and subsequent interpretation accuracy reduction [4]. Therefore, it becomes imperative to explore and develop effective methods for range ambiguity suppression.

B. Related Works

In recent years, the primary approaches for range ambiguity suppression in spaceborne SAR have originated either at the transceiver system level or the post-processing level [5].

1) The transceiver system method predominantly involves the encoding or design of the transmitted waveform. Azimuth phase coding (APC) technology serves to distinguish between the desired signals and range ambiguity in single-channel SAR systems. It employs azimuth phase modulation on the transmitting pulse and corresponding demodulation, resulting in a spectral shift of the range ambiguity in the Doppler domain, allowing it to be filtered out [6]. However, in the azimuth multi-channel SAR system, it will lead to the Doppler spectrum folding due to the system's low pulse repetition frequency (PRF).

Krieger et al. [7] proposed another significant technique: multidimensional waveform coding coupled with digital beamforming (DBF). By exploiting differences in the direction of arrival, it utilizes a weighted sum to eliminate range ambiguity stemming from undesired signal directions. Subsequently, a reconstruction technique based on minimum variance distortion response (MVDR) beamforming was proposed to remove the range ambiguity in both the range and azimuth time domain [8]. Likewise, sophisticated schemes grounded in convex optimization have been suggested for multi-channel systems [9], [10]. The performance of these DBF technologies relies on spatial degrees of freedom. To address this challenge, the frequency diversity array (FDA) has been employed to improve the flexibility of the system in the multiple input multiple output radar [11], [12].

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In contrast to the aforementioned technologies, another approach involves the design of the transmission waveform. Specifically, the up-and-down chirp (UDC) modulation is proposed for range ambiguity suppression. In this system, the echo originating from the first ambiguity regions bears an opposite chirp rate to the reference signal, resulting in a low gain through the matched filter [13]. Guo et al. [14] devised an orthogonal nonlinear frequency modulation (NLFM) signal based on Doppler division multiple access (DDMA) for single-channel spaceborne SAR imaging. This design effectively elevates the degrees of freedom and enhances the range ambiguity signal ratio (RASR) [14]. Jin et al. [15] proposed a novel range azimuth joint modulation transmission scheme and designed a NLFM waveform with continuous segmented linear instantaneous frequency. Juxtaposed with the APC, a waveform with low cross correlation energy is proposed, elevating the additional range sampling margin [16]. Xu et al. [17] proposed an orthogonal NLFM signal aimed at simultaneously mitigating spaceborne SAR's sidelobe range ambiguity and improving range resolution [16]. Nevertheless, a majority of these methods remain in the simulation stage.

2) In addition to transceiver system processing, range ambiguity can also be mitigated during post-processing. Under the UDC modulation system, a dual focus method is proposed to remove the range ambiguity or nadir echo – a special type of range ambiguity. This method has three steps: range ambiguity focusing by redesigning matching filters, range ambiguity detection by constant false alarm detection (CFAR), the desired signal focusing by inverse focusing, and secondary imaging. It ensures the elimination of the range ambiguity without seriously damaging the desired signal [4], [18]. Nevertheless, this method requires considerable time and resources for the last step. To overcome this problem, a novel end-to-end method called delta focusing is proposed. This method enables imaging straightly after nadir echo suppression rather than inverse focusing and secondary imaging [19], which halved the computational burden approximately. Recently, a range ambiguity suppression method combining matching filters and sparse reconstruction has been proposed [20]. It uses data from the range ambiguity regions to estimate and reconstruct the range ambiguity, ultimately obtaining a non-ambiguous image through echo cancellation and matched filtering. However, it is not suitable for high-squint scenarios.

Existing post-processing methods have shown promising results in the raw echo domain. However, most users usually only obtain single-look-complex (SLC) images instead of raw echo. So, it is difficult to remove the range ambiguity on SLC images for current methods not focusing on image domain processing. Furthermore, some ghost artifacts on SLC images exhibit characteristics similar to range ambiguity [21]. Traditionally, identifying range ambiguity involves delaying or advancing the receiving time of the raw echo by N PRIs to observe artifact focusing. This makes it a challenge for users to discern artifact types on SLC images. Therefore, it is necessary to design a method meeting the needs of users to identify and remove range ambiguity on SLC images quickly and accurately.

C. Contributions

This article relies on range ambiguity recognition and detection in spaceborne SAR systems to efficiently and effectively repair the SLC product affected by range ambiguity. The main contributions are summarized as follows.

1) The multi-dimensional representation of range ambiguity in the image domain is simulated and analyzed. The distinct disparities of characteristics between the range ambiguity and desired signal are revealed in the azimuth fractional Fourier transform (FRFT) domain.
2) This article introduces a new concept about range ambiguity recognition and suppression in the image domain, rather than in the echo domain. A method via FRFT and generalized extreme studied deviate (GESD) is proposed to identify and remove the strong range ambiguity from the SLC images without prior knowledge.
3) The typical real measured images with range ambiguity and their experiment results of the spaceborne SAR system are provided. The result shows that the range ambiguity could be identified and removed by the proposed method effectively. Additionally, quantitative analysis shows a notable decrease in RASR, objectively highlighting the effectiveness of the method.

The remainder of this article is organized as follows. The model of range ambiguity is introduced in Section II. Section III presented the proposed methods in detail. Section IV provides experimental results on the real measured SLC data of typical scenarios, while Section V concludes the article.

II. Problem Formulation

A. Range Ambiguity Model

Here, for the convenience of analysis of the range ambiguity imaging phenomenon, the observation geometry view is built as straight-line as an approximation of Earth shown in Fig. 1. Fig. 1(a) is a sidelobe range ambiguity model and Fig. 1(b) is the cross-sectional drawing of it. The signal source satellite over the shortest range R_0 observes the Earth by transmitting and receiving electromagnetic waves to the ground at an azimuthal velocity of V_r. The range ambiguity from the sidelobe of the ambiguity region at the previous PRI and the desired signal from the main lobe of the observed region at the current time are received at the same time after N PRIs, represented by red dashed lines. X, X_r, θ_{sq}, R, R_r and R_{ra} are the desired signal ground range, the range ambiguity ground range, the desired signal squint angle, the desired signal slant range, the range ambiguity slant range, and the ground range in one PRI respectively.

Then, the total received echo $\mathbf{s}_E \in \mathbb{C}^{N_r \times N_a}$ is expressed as (1)

$$\boldsymbol{s}_E(\tau, \eta) = s_r(\tau, \eta) + s_{\text{ra}}(\tau, \eta) + s_{\text{noise}}(\tau, \eta) \tag{1}$$

where $s_r \in \mathbb{C}^{N_r \times N_a}$ represents the desired signal, $s_{\text{ra}} \in \mathbb{C}^{N_r \times N_a}$ represents the range ambiguity, $s_{\text{noise}} \in \mathbb{C}^{N_r \times N_a}$ represents the noise signal. N_r and N_a represent the range and azimuth sample points respectively.

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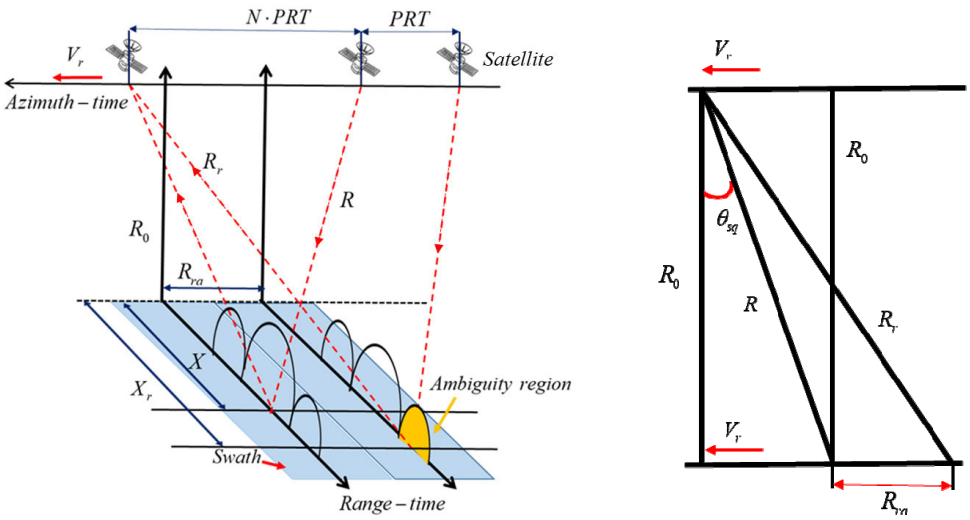


Fig. 1. Principle of range ambiguity. (a) Approximation of the straight-line geometry to the curved geometry of the Earth. (b) Cross-sectional drawings.

If θ_{sq} are not considered, the ideal echo model of a point target is [22]

$$s_r(\tau,\eta) = A_r\omega_a(\eta-\eta_c)\omega_r\left(\tau-\frac{2R(\eta)}{c}\right) \exp\left(-j\frac{4\pi R(\eta)}{\lambda}\right)\exp\left(j\pi K_r\left(\tau-\frac{2R(\eta)}{c}\right)^2\right) \quad (2)$$

where A_r is the amplitude of the echo, τ represents the range fast time, η represents the azimuth slow time, λ represents the radar wavelength, $\omega_r(\cdot)$ represents the envelope of the desired signal in the range time domain, $\omega_a(\cdot)$ represents the beam pattern in the azimuth domain of receive echo, K_r is the range chirp rate.

Assuming that for a certain PRT moment, both the current PRT and the previous PRT echo are received at the same time, then range ambiguity will occur, and the echo model of the range ambiguity can be expressed as the following equation:

$$s_{\text{ra}}(\tau,\eta) = A_{\text{ra}}\omega_a(\eta-\eta_c)\omega_r\left(\tau-\frac{2R_r(\eta)}{c}\right) \exp\left(-j\frac{4\pi R_r(\eta)}{\lambda}\right)\exp\left(j\pi K_r\left(\tau-\frac{2R_r(\eta)}{c}\right)^2\right) \quad (3)$$

where $R_r(\eta)$ denotes $(R_0^2+X_r^2+V_r^2(\eta+\text{PRT})^2)^{1/2}$. A_{ra} is the amplitude of the range ambiguity.

After range compression, range cell migration correction (RCMC), and azimuth compression, the expression of the desired non-ambiguity signal is the following equation:

$$s_r(\tau,\eta) = \text{Ap}_a(\eta)p_r\left(\tau-\frac{2R_c}{c}\right) \times \exp\left(-j\frac{4\pi R_c}{\lambda}\right)\exp\left(j2\pi f_{\eta c}\eta\right) \quad (4)$$

where $f_{\eta c}$ is the Doppler center frequency, R_c is the constant term $(R_0^2+X^2)^{1/2}$. $p_a(\eta)$ and $p_r(\tau-(2R_c/c))$ is the envelope in the form of Sinc$(\cdot)$, while the point target is at $\eta=0, \tau=(2R_c/c)$.

Similarly, the expression of range ambiguity can be expressed as (5). The detailed derivation process from (3) to (5) is shown in Appendix A

$$s_{\text{ra}}(\tau,\eta) = A_{\text{ra}}A_1 p_a(\eta)p_r\left(\tau-\frac{2\text{PRT}}{c}\sqrt{\frac{R_c^2}{\eta^2}+V_r^2}\right) \exp\left(-j\frac{4\pi R_{ca}}{\lambda}-j\pi K_{ar}\text{PRT}^2\left(1+\frac{R_{\text{ra}}^2}{2R_{ca}^2}\right)\right) \exp\left(-j\pi K_{ar}\left(\frac{R_{\text{ra}}^2}{2R_{ca}^2}\right)\eta^2 - j2\pi\eta\left(K_{ar}\text{PRT}\left(1+\frac{R_{\text{ra}}^2}{2R_{ca}^2}\right)+f_{\eta c}\right)\right) \quad (5)$$

where K_{ar} represents the range ambiguity azimuth chirp rate, which equals to $(2V_r^2/\lambda(R_{ca}+(R_{\text{ra}}^2/2R_{ca})))$, R_{ca} is the constant term $(R_0^2+X_r^2)^{1/2}$, and A_1 is defined as $\exp(-j(4\pi\text{PRT}/\lambda))$. If the value of PRT and R_{ra} are ignored, (5) can be simplified as (4).

Compared to (4), there are incorrect parameters causing deviation in RCMC, defocused energy response in azimuth compression, and a linear shift in the envelope $p_r(\cdot)$ concerning azimuth time η in (5). Meanwhile, if the point target is extended to an area target, it manifests as multiple bright lines parallel along the azimuth and covers the desired target. Finally, according to the analysis by (5), it is shown that the phase of range ambiguity is a linear frequency modulation form that varies with the azimuth time η. Compared with the non-ambiguity signal model (4), the range ambiguity can be separated from the azimuth spectrogram domain.

B. Simulation Spectrum Analysis

Based on the above model, this article carried out a point target simulation under the low squint angle to roughly explain the characteristics of range ambiguity. The main system parameters are summarized in Table I.

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TABLE I
PARAMETERS OF THE SIMULATION SAR SYSTEM

Parameters	Value
PRF	1451Hz
Wavelength	0.2381m
Bandwidth	80MHz
Sample Frequency	90MHz
Pulse Width	20μs
Carrier Frequency	1260MHz
Azimuth Aperture Length	9.6m
Look Angle	10°
Platform Velocity	7300m/s
Center Slant range	800km
Near Slant range	793km
Far Slant range	810km
Point Target	(500,0,0)

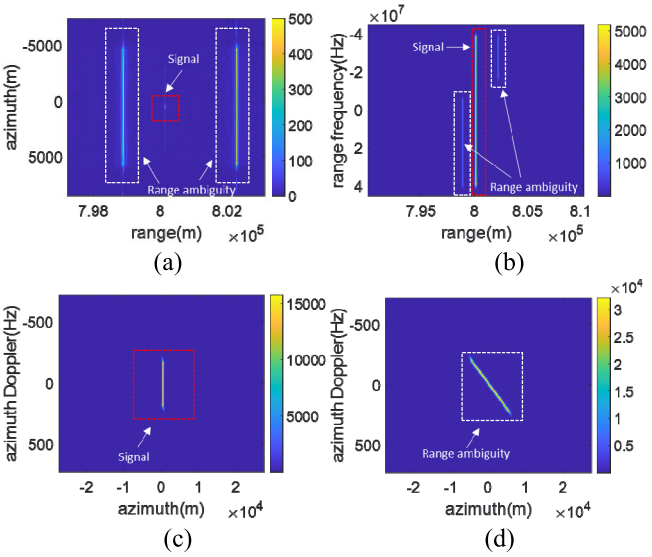


Fig. 2. Range ambiguity simulation. (a) Point target and range ambiguity imaging. (b) Point target and range ambiguity in the range spectrogram domain. (c) Point target in azimuth spectrogram domain. (d) Range ambiguity in the azimuth spectrogram domain.

Fig. 2(a) shows the region of interest (ROI) of the range-Doppler algorithm (RDA) for 1-point target and 2 range ambiguities. The red ROI shows the focusing point target, while the white ROIs show the range ambiguities with a vertical shape. This shape matches the envelope $p_r(\cdot)$ with the linear shift analyzed in Section II-A.

Then, extract the pulse of the point target at an azimuth of 0 m and observe its range spectrogram, as shown in Fig. 2(b). Both range ambiguity and desired signal are compressed into vertical lines for their same range chirp rate. It is impossible to distinguish them in the range spectrogram. Fig. 2(c) shows the azimuth spectrogram from the range cell of the point target of the red ROIs. The point target is compressed into a vertical line parallel to the azimuth. Similarly, extract the range cell of the range ambiguity of the left white ROIs in Fig. 2(a). The range ambiguity appears as a skewed-shape line, as shown in Fig. 2(d). It is consistent with (4) and (5) that the frequency changes with time linearly in the azimuth spectrogram.

C. Measured Data Spectrum Analysis

To further explore the characteristics of the range ambiguity in the image domain, the article provides an example scenario collected by a spaceborne L-band SAR and analyzes its spectrum. The distorted example scenario A was obtained over Sichuan, Province, China with a size of 16 384 × 40 106. The red ROIs of Fig. 3(a) show the range ambiguity is composed of a large number of white thin lines, indicating azimuth defocus in the measured data. Fig. 3(b) and (c) are the range spectrogram and the azimuth spectrogram derived from the 6001st pulse and the 1412th range cell of scenario A respectively. In Fig. 3(b), range ambiguity is visible within the white ROIs, and its energy is marginally stronger than the desired signal. However, the time-frequency concentration of the range ambiguity in the range spectrogram is not ideal. The irregular shape of the range ambiguity also increases the difficulty of mitigation. In contrast, in Fig. 3(c), the time-frequency concentration of the range ambiguity in the azimuth spectrogram is superior to that in the range spectrogram.

Fig. 3(d) illustrates that the amplitude of the range ambiguity in black ROIs is higher than the desired signals, except for the red ROIs. So, the first idea is to convert 1-D data into 2-D data by short-time Fourier transform (STFT), protect the strong scattering points, and mitigate the strong range ambiguity. However, the idea has two obvious limitations. The first one is the selection of thresholds. A single hard threshold makes distinguishing between the range ambiguity and desired signals difficult. Another limitation relates to computational complexity. It has a large quantity of calculations about detecting range ambiguity point by point in a 2-D space.

The second idea is drawing inspiration from the traditional identifying range ambiguity method. Fig. 4 shows the 1412th range cell in the azimuth spectrogram after delaying one PRI and imaging. The range ambiguity is focused while the scattering point is defocused, which is opposite to the phenomenon in Fig. 3(c).

The operation of delaying one PRI and imaging manifests as the rotation of the time-frequency space in Fig. 4. So, if the skewed-shape range ambiguity is rotated from Figs. 3(c) and 4, it will also focus. The focus enables a direct determination of whether the artifacts are due to the range ambiguity. This rotation operation can be implemented by FRFT. Additionally, when the range ambiguity is focused, its energy accumulates coherently, whereas the energy of the desired signals becomes dispersed in the transformed domain. This makes it easier to detect the range ambiguity compared to the desired signals.

Based on the above analysis, the flowchart of the proposed method is shown in Fig. 5. The first step is estimating the order of spatial transformation performed on the image through preprocessing α. Subsequently, an α order FRFT transform is executed to detect outliers in a new transform domain. Then, a notched filter is applied to these outliers. At last, a $-\alpha$ order FRFT transformation is performed to revert back to the original focused SLC image. In Section III, the article will elaborate on the principles behind each step and justify their usage.

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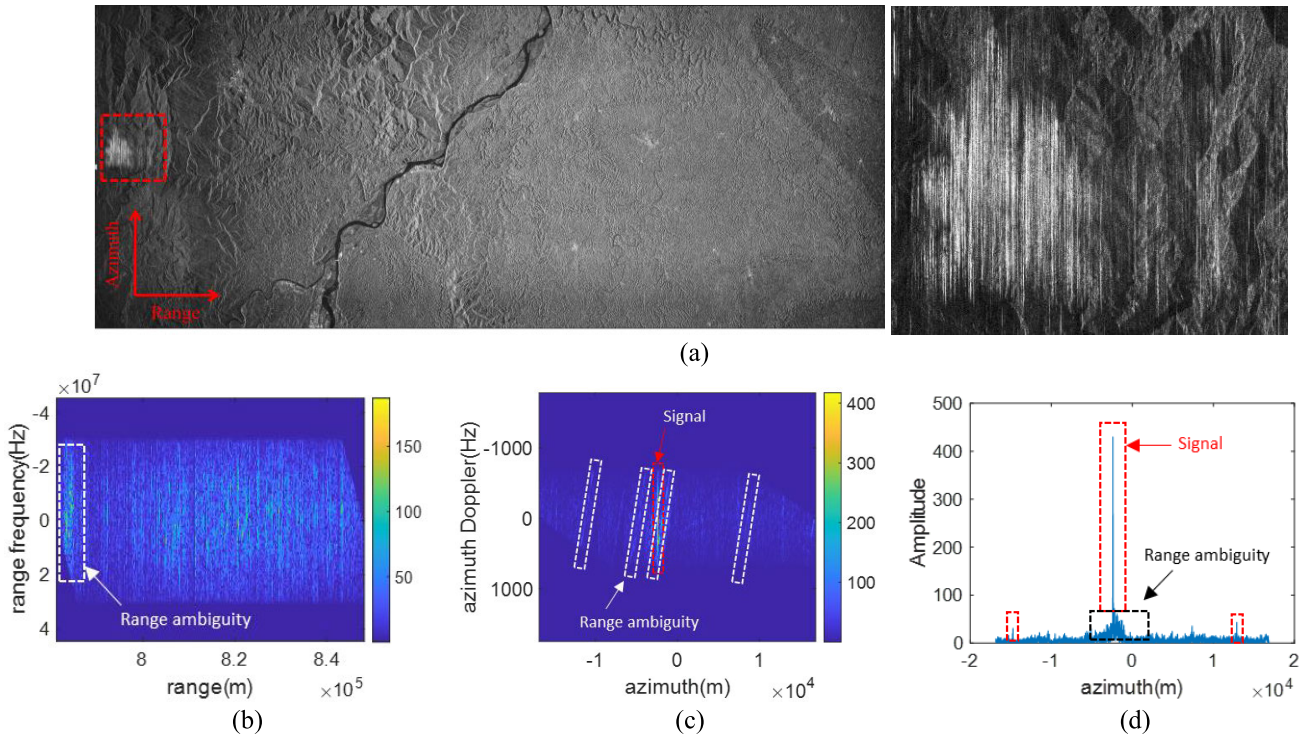


Fig. 3. Phenomenon of range ambiguity in scenario A. (a) Original image and its red ROI. (b) Imaging in the range spectrogram domain. (c) Imaging in azimuth spectrogram domain. (d) 1412th range cell of scenario A.

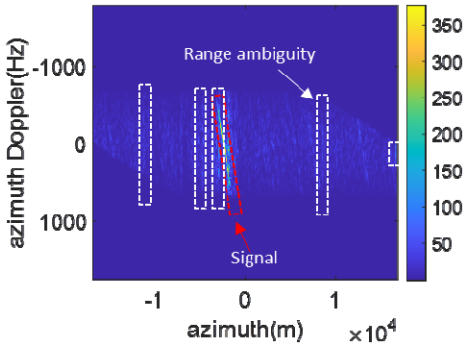


Fig. 4. Mismatched imaging in the azimuth spectrogram domain.

III. Theory and Methodology

A. Preprocessing

This section introduces the identification and suppression method proposed in the article in more detail. To completely separate range ambiguity, FRFT is used to maximize the difference in energy aggregation between range ambiguity and the desired signal in the spectrogram. FRFT is a classic method which is often used in linear frequency modulation parameter estimation [23], [24]. It can overcome the time-frequency resolution constraints by projecting the data onto a linear frequency modulation basis. Here, it is used to construct a maximum separable space between the desired signal and range ambiguity for subsequent detection. The FRFT can be expressed as

$$X_\alpha(u) = F_\alpha(x(t)) = \int_{-\infty}^{+\infty} x(t) K_\alpha(t,u)dt \tag{6}$$

where the $x(\cdot) \in \mathbb{C}^{N_r \times 1}$ is a range cell of the SLC image $x(\tau, t) \in \mathbb{C}^{N_r \times N_a}$, $F_\alpha(\cdot)$ is the α-order FRFT, u is the FRFT domain, K_α is the transform kernel. $X_\alpha(\cdot) \in \mathbb{C}^{N_r \times 1}$ represents the SLC data after α-order FRFT.

The accuracy of FRFT order α is pivotal for effective range ambiguity detection and suppression. The traditional method of searching FRFT order has limitations in the convergence speed and accuracy in processing data with multiple frequency modulation rates. However, according to (5), the azimuth chirp rate of all range ambiguities is nearly constant. Consequently, the estimation obtained from one range cell can be extrapolated to other range cells.

According to the analysis in Section II, when the range ambiguity is correctly focused, it is shown as narrowband signals with sufficient sparsity in the α-transform spectrogram and maximum amplitude. Considering the strong sparsity, the $L0$-norm and maximum amplitude are jointly used as the search condition for compactness, which can be written as

$$\alpha_{\text{opt}} = \arg\min(\|X_\alpha(u)\|_0 - R(X_\alpha(u))) \tag{7}$$

where α is the order of transformation, u is the pulse number and $R(\cdot)$ is the absolute value of the maximum value among $X_\alpha(\cdot)$.

$L0$-norm is defined as the number of non-zero sample points in a vector, which is easily affected by low noise, making the optimization process an NP-hard problem. Numerous studies have shown that the $L1$-norm is a convex approximation of the $L0$-norm. Therefore, the $L1$-norm is used as the search constraint. Equation (7) can be written as

$$\alpha_{\text{opt}} = \arg\min(\|X_\alpha(u)\|_1 - R(X_\alpha(u))). \tag{8}$$

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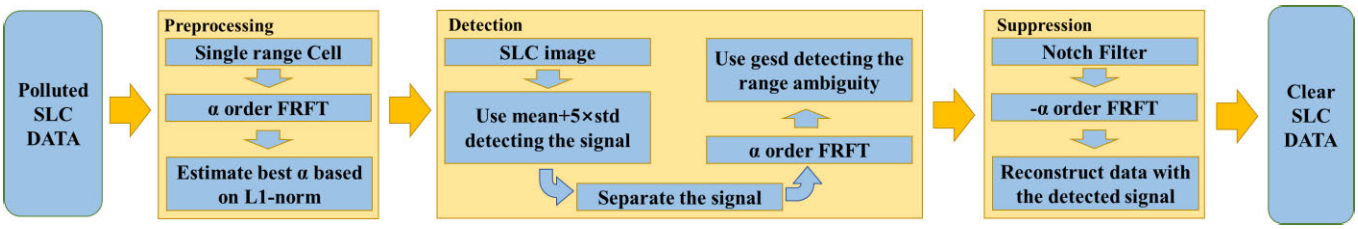


Fig. 5. Workflow of the proposed method.

However, the maximum value $R(\cdot)$ may not be robust in all situations. If the energy of the desired signal is only slightly higher or lower than the range ambiguity, the $L1$-norm will serve as a constraining factor, helping to pinpoint the true α value. Conversely, if the energy of the strong scattering point is significantly higher than the surrounding pulses like Fig. 3(d), the α estimation results based on (8) may have errors. At this point, it becomes necessary to eliminate the extremely strong scattering point by setting a relatively high threshold. Here two hypotheses H_0 and H_1 are established, which are shown in the following equation:

$$\begin{cases} H_0 : |X_\alpha(u_i)| \le \text{Th} \\ H_1 : |X_\alpha(u_i)| > \text{Th} \end{cases} \tag{9}$$

where the threshold Th is designed as

$$\text{Th} = \text{mean}(|X_\alpha(u)|) + 5 \times \text{std}(|X_\alpha(u)|). \tag{10}$$

The hypotheses indicate that if a point u_i in $X_\alpha(u)$ is less than or equal to the threshold Th, this point is not a strong scattering point. If a point u_i in $X_\alpha(u)$ is higher than the threshold Th, it needs to be removed. Due to the characteristic of range ambiguity as a wideband signal in the azimuth time domain, its energy is dispersed across various pulses, so threshold detection will not cause false positives.

Then, the implemented algorithm to search α is given in Algorithm 1, where α_{opt} is the optimal α, κ is the search interval and ε is the termination condition. Fig. 6 shows the normalization results of the minimum objective optimization function under the different α-transformation, where the blue line is the optimization objective function proposed in this article, red is the $L1$-norm-based processing method proposed in [25], and yellow is the supplementary convex optimization function. Obviously, due to the particularly low energy of the out-of-band signal in the azimuth frequency domain, its transform domain is relatively sparse. It will cause incorrect search results only using the $L1$-norm as a constraint in the azimuth dimension. Finally, Fig. 7(a) shows the result of the 1412th range cell after FRFT, where range ambiguity is concentrated into lines. Fig. 7(b) is the α-transform spectrogram corresponding to Fig. 7(a), demonstrating that the range ambiguity has been focused from another perspective.

Algorithm 1 Optimal α Search

Input: $x(t)$
Initialization: $\alpha_{begin} = 0, \alpha_{end} = 4, \Delta\alpha = 0.1$
delete outlier by (6) ***and*** (6))
  While $\Delta\alpha < \varepsilon$ ***do***
    for $\alpha = \alpha_{begin}$***to***α_{end}***step***$\Delta\alpha$ ***do***
      calculated $\|X_\alpha(u)\|_1 - R(X_\alpha(u))$ by (6);
    end for
    $\alpha_{\text{opt}} = \arg\min(\|X_\alpha(u)\|_1 - R(X_\alpha(u)))$
    $\alpha_{begin} = \max\{\alpha_{begin}, a_{\text{opt}} - 0.5\Delta a\}$
    $\alpha_{end} = \max\{\alpha_{end}, a_{\text{opt}} - 0.5\Delta a\}$
    $\Delta\alpha = \Delta\alpha \times \kappa$
  end while
Output α_{opt}

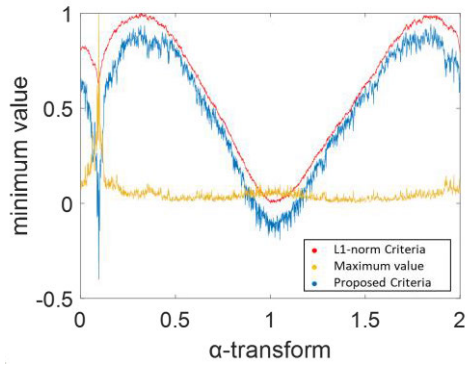


Fig. 6. Minimize search best α under different criteria.

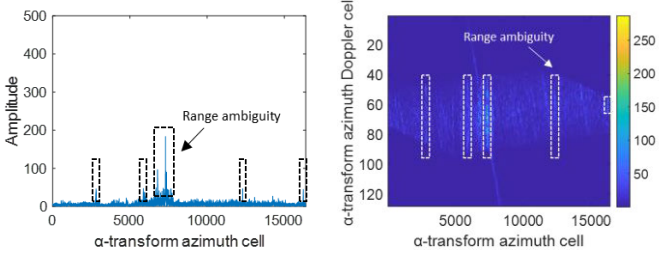


Fig. 7. α-transform domain of scenario A. (a) Range cell with range ambiguity after FRFT. (b) Range cell α-transform spectrogram with range ambiguity after FRFT.

B. Ambiguity Identify and Detection

Before executing azimuth FRFT to all the range cells, it is necessary to take the same operation as (10) to remove the extremely strong scattering point. Then, all range units of the SLC image will be processed by α order FRFT. The process above can be written as

$$\begin{aligned} X_\alpha^*(t,u) &= F_\alpha(x(\tau,t)) \\ &= \int_{-\infty}^{+\infty} (x(\tau,t) - x_s(\tau,t)) K_\alpha(\tau,t,u) d\tau \end{aligned} \tag{11}$$

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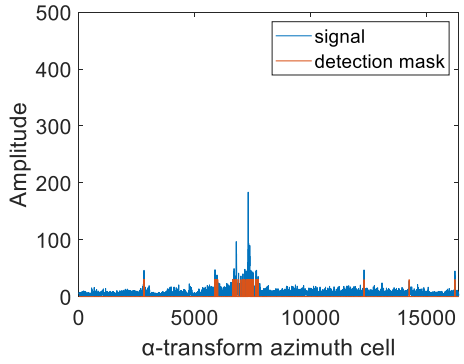


Fig. 8. Detection results by GESD.

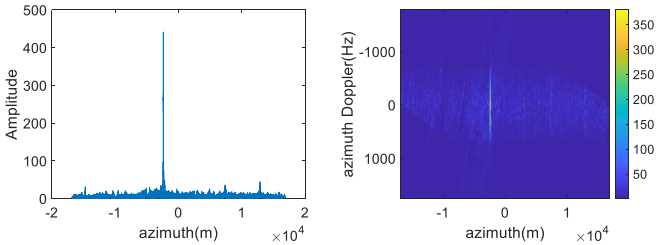


Fig. 9. Suppression results by notched filter and IFRFT. (a) Range cell after range ambiguity suppression and reconstruction. (b) Range cell azimuth spectrogram after range ambiguity suppression and reconstruction.

where the $x_s(\tau, t) \in \mathbb{C}^{N_r \times N_\alpha}$ is the strong scattering point. $X_\alpha^*(\cdot) \in \mathbb{C}^{N_r \times N_\alpha}$ denotes the SLC data after α-order FRFT.

After executing FRFT, the range ambiguity should be detected in the transformation domain. For this purpose, GESD is used for identifying discrete values [26]. Outlier detection for 1-D data has high efficiency. The idea of identifying anomalies by sorting the observed data according to the size of the deviation center is to detect multiple outliers. The specific step is to compare the statistic R_j with the critical value ψ_j, if $R_j > \psi_j$, this point is an outlier, where R_j and ψ_j is represented as

$$R_i = \frac{|x_i - \bar{x}|}{\delta} \tag{12}$$

$$\psi_j = \frac{(n-1)t_{n-i+1,p}}{\sqrt{(n-i-1)\left(n-i-1-t_{n-i+1,p}^2\right)}} \tag{13}$$

where the $\bar{x}$ is the mean of $X_\alpha(t, u)$, δ is the standard deviation, ψ_j is the critical value of the top j largest numbers, n is the number of discrete points, $t_{n-i+1,p}$ is the value of the right tail probability p for the t-distribution of $n-i+1$ degrees of freedom.

Fig. 8 shows the detection results of Fig. 7(a). The red cells are judged as range ambiguity. In addition to detecting points with higher peaks, GESD also detects some weak range ambiguity in the transformation domain.

C. Suppression and Reconstruction

In this section, the frequency notch filter (FNF) will be used for suppression after range ambiguity detection. FNF is a classic incoherent suppression method [27]. Thanks to the dispersion of the desired signal energy by FRFT, the shortcoming of signal loss caused by the FNF is minimized as much as possible.

Based on the outlier detection results by GESD, the notched filter $\text{Mask}(t, u) \in \mathbb{C}^{N_r \times N_\alpha}$ is defined as

$$\text{Mask}(t, u) = \begin{cases} 1, & X_\alpha(t, u) \in \Omega \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where the Ω is the set of outliers that satisfies $R_j > \psi_j$.

Then, the desired signal $X_\alpha(t, u)$ is orthogonally projected onto the notched filter $\text{Mask}(t, u)$ for range ambiguity suppression. Next, -α order FRFT will be taken to reconstruct the image. Finally, the strong scattering point removed in preprocessing will be added to the image. The whole suppression and reconstruction operation are as follows:

$$x^*(\tau, t) = F_{-\alpha}((1 - \text{Mask}(t, u)) \times X_\alpha(t, u)) + x_s(\tau, t) \tag{15}$$

where $x^*(\tau, t) \in \mathbb{C}^{N_r \times N_\alpha}$ is the recovery SLC image.

Fig. 9 shows the outcome of the range cell and azimuth spectrogram. Compared with Fig. 3(d), the range ambiguity is removed while the strong scattering point is retained.

TABLE II
PARAMETERS OF THE MEASURED SAR SYSTEM

Parameters	Scenario A	Scenario B
PRF	3534Hz	2902Hz
Wavelength	0.2381m	0.2381m
Bandwidth	60MHz	80MHz
Sample Frequency	90MHz	90MHz
Carrier Frequency	1260MHz	1260MHz

IV. EXPERIMENTAL RESULTS

A. Experimental Settings

In this section, real measured SLC products are set for performance validation. Two experiment scenarios, including scenario A, were both collected from the L-band spaceborne interferometry SAR in strip mode and fixed chirp modulation system. The main system parameters are listed in Table II. Four other methods, named autocorrelation (AC) [21], 2D-CFAR [4], moving window (MW) [18], and proposed STFT, are used as a scientific control for the proposed FRFT.

RASR and time will be used as performance evaluation indicators. The RASR is defined as the ratio of the total energy in the range ambiguity regions to the total energy in all wide swaths

$$\text{RASR} = \frac{\sum_{n=-\infty, n\neq 0}^{\infty} \int_{\frac{-B_p}{2}}^{\frac{B_p}{2}} S(f, \tau + n \cdot \text{PRT}) df}{\int_{\frac{-B_p}{2}}^{\frac{B_p}{2}} S(f, \tau) df} \tag{16}$$

where $S(f, \tau)$ is the energy of the desired signal. $S(f, \tau + n \cdot \text{PRT})$ is the energy of range ambiguity regions. B_p represents the Doppler bandwidth. f represents the Doppler frequency. In this article, all $S(f, \tau)$ and $S(f, \tau + n \cdot \text{PRT})$ have been normalized.

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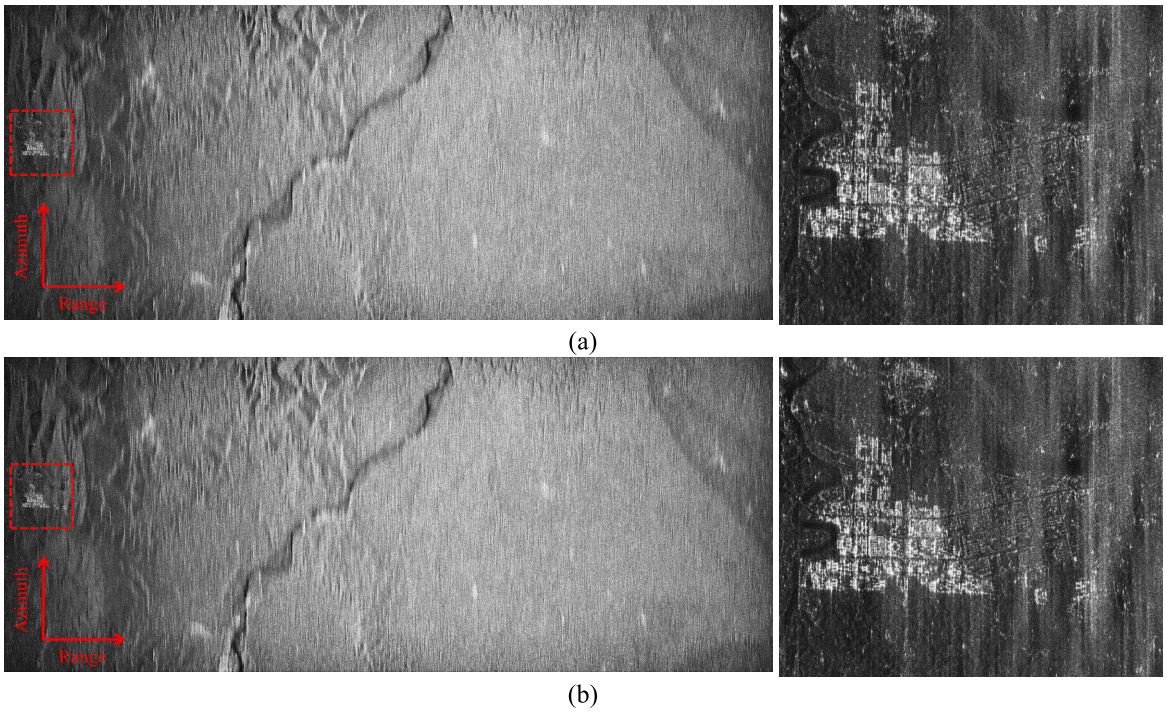


Fig. 10. Focusing result of range ambiguity in scenario A. (a) Mismatched imaging by delaying PRI and its red ROI. (b) Mismatched imaging by FRFT and its red ROI.

B. Scenario A

Fig. 10(a) shows the results of mismatched imaging by inverse focusing on the SLC data and delaying it by one PRI. The white thin lines are focused on the country, river, and land. This shows that the lines are the range ambiguity. Fig. 10(b) shows the results of imaging through FRFT on the SLC data. Compared with Fig. 10(a), the white lines are also focused, which indicates that the range ambiguity types can be directly recognized in the SLC data by the proposed method.

Fig. 11 shows the suppression results of five methods for scenario A and their ROI. Fig. 11(a) shows that the ROI based on AC leaves many residues. The main reason is the range ambiguity in scenario A lacks obvious characteristics in the range-Doppler (RD) domain. Fig. 11(b) and (c) are the methods of 2D-CFAR and MW respectively. Both methods are employed in the UDC modulation system. The primary distinction between the two methods lies in the construction of the detection window and threshold setting. 2D-CFAR and MW still take effect in range ambiguity suppression under a fixed chirp modulation system, while MW has a better performance than 2D-CFAR. Fig. 11(d) shows there are still a few range ambiguities using the proposed STFT. Fig. 11(e) shows the performance of the proposed FRFT is as good as the MW. In summary, the results of scenario A preliminarily highlight the utility of the proposed approaches in enhancing radar imaging quality by reducing artifacts caused by range ambiguities.

C. Scenario B

Scenario B was distortion obtained over Inner Mongolia Province, China with a size of 16 384 × 17 998. Compared to scenario A, vertical bright range ambiguities are distributed in large areas shown in the green ROIs of Fig. 12(a). Partial range ambiguities appear in multiple urban areas, increasing detection difficulty. After mismatched imaging, the range ambiguities from the red ROI are focused on highways, overpasses, or the other country shown in Fig. 12(b), while the original country in Fig. 12(a) appears defocused. Similar to scenario A, the ambiguous regions are often from the strong scatter targets such as cities and railways. The gain of coherence in pulse compression systems results in the energy of range ambiguity of other characteristic types weaker than the original. Fig. 12(c) shows the imaging results through performing FRFT on the SLC data, demonstrating the feasibility of the proposed method for identifying the range ambiguity type. Compared to Fig. 12(b), the concentration of range ambiguity by FRFT is not as good as parameterization processing. This is due to the difference in azimuth chirp rate K_{ar} between the near and far slant ranges. The optimal α estimation via a single range cell will have deviations not exceeding 0.01. Therefore, if the observation scene is too large, the focusing performance of the range ambiguity is slightly inferior.

Fig. 13 shows the experimental results of scenario B processing results. The right image represents the ROI of the red box in the left image. Fig. 13(a) shows the method of AC suppression result. AC is used to remove a type of ghost artifact that has similar characteristics to range ambiguity on SLC images. Combined with the scenario A results, AC is unsuitable for range ambiguity suppression. This illustrates the importance of identifying the ghost artifact type in advance. Fig. 13(b) shows that the method of 2D-CFAR has some false alarms, while Fig. 13(c) shows that the method of MW has the desired signal loss in the ROI of the red and green box in the left image. Fig. 13(d) shows the relatively dark

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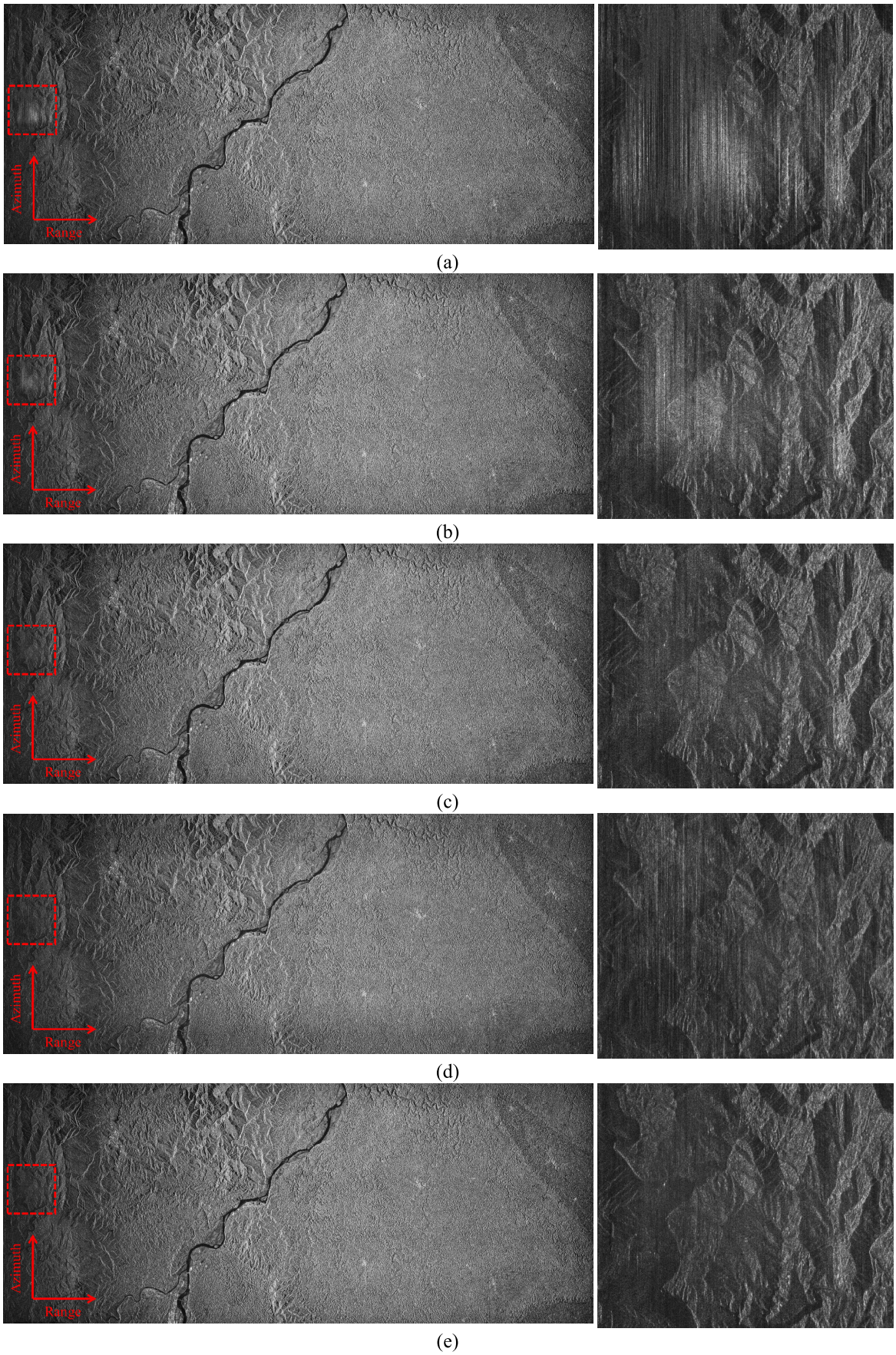


Fig. 11. Suppression result in scenario A. (a) AC [21]. (b) 2D-CFAR [4]. (c) MW [18]. (d) Proposed STFT. (e) Proposed FRFT.

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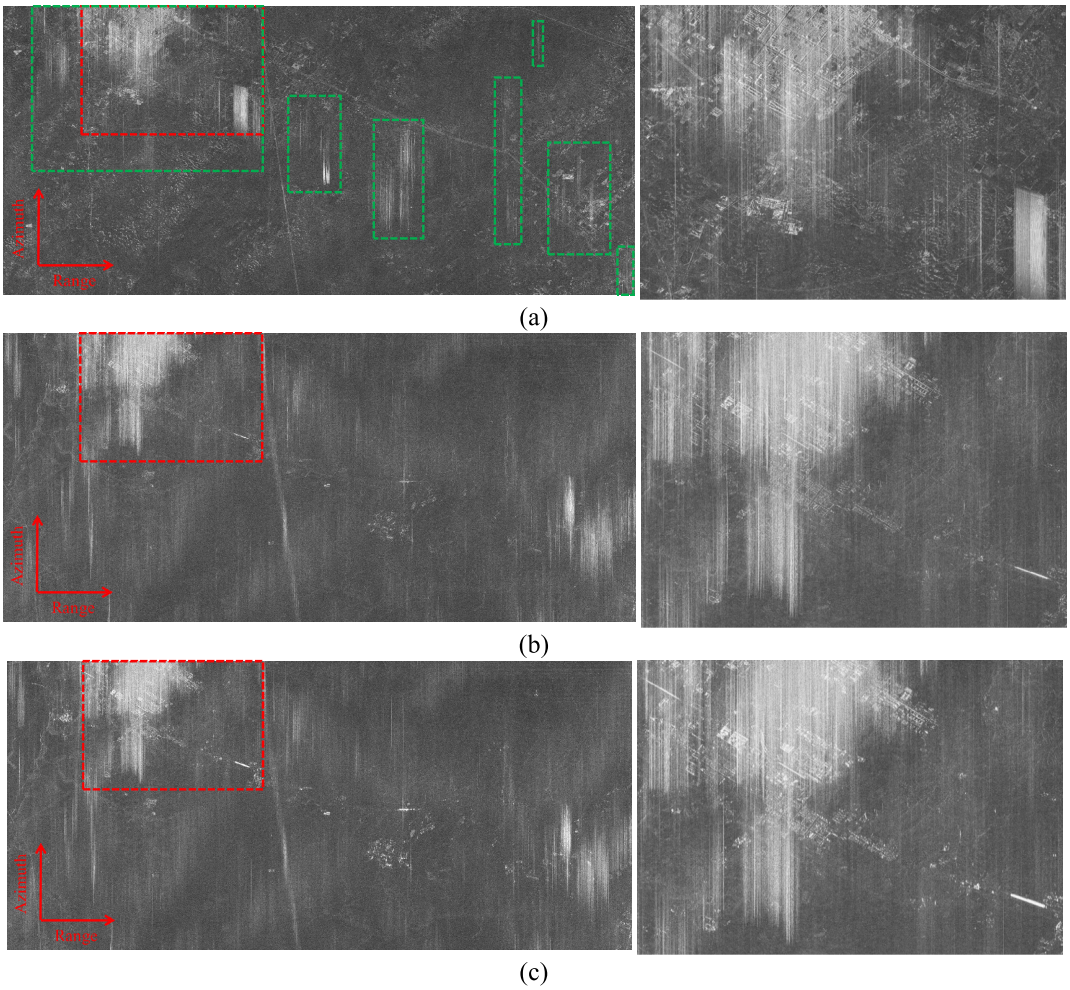


Fig. 12. Phenomenon of range ambiguity in scenario B. (a) Original image and its red ROIs. (b) Mismatched imaging by delaying PRI and its red ROIs. (c) Mismatched imaging by FRFT and its red ROIs.

scene after STFT, which has some false alarms and the loss of desired signals. It is because the detector of STFT only considers the energy but ignores the characteristics difference between the range ambiguity and desired signals. Fig. 13(e) shows the range ambiguity suppression performance of the proposed FRFT is as good as MW, but avoids the loss of desired signals.

D. Quantitative Result Analysis

Table III compares RASR across five methods. It is shown that the method of AC has the worst RASR while the proposed FRFT has the best in both scenarios, which is consistent with the qualitative results in Sections IV-B and IV-C.

Table III also shows the processing times for the five methods. Among them, in scenario A, the proposed STFT and FRFT only require processing of the range cells located within the ambiguity regions. Conversely, in scenario B, all methods address the entire panorama. The method of AC involves nine autocorrelation calculations and comparisons for each point in the RD domain, making it impractical for processing in large-scale scenes. For 2D-CFAR and MW, the main usage time is twice inverse focusing and twice imaging on the SLC data. The computational complexity of this process is

$$T = 4O\left(2N_aN_r\log_2 N_r + 2N_aN_r\log_2 N_a + 5N_aN_r\right) \quad (17)$$

TABLE III
PERFORMANCE COMPARISON OF THE DIFFERENT SCENARIOS

Scene	RASR (Origin)	Method	RASR (Filter)	Time
A	-2.51dB	AC [21]	-4.36dB	406.89s
		2D-CFAR [4]	-5.17dB	631.03s
		MW [18]	-6.07dB	941.27s
		Proposed STFT	-6.61dB	246.77s
		Proposed FRFT	-6.99dB	92.15s
B	-7.97dB	AC [21]	-9.19dB	8676.60s
		2D-CFAR [4]	-12.21dB	803.46s
		MW [18]	-12.49dB	1232.00s
		Proposed STFT	-12.13dB	1992.98s
		Proposed FRFT	-12.82dB	743.14s

where N_a is the pulse number, N_r is the range cell number.

The computational complexity of GESD is $O(logN)$. The computational complexity of a single STFT or FRFT is $O(logN)$. However, during the detection process, the SLC data

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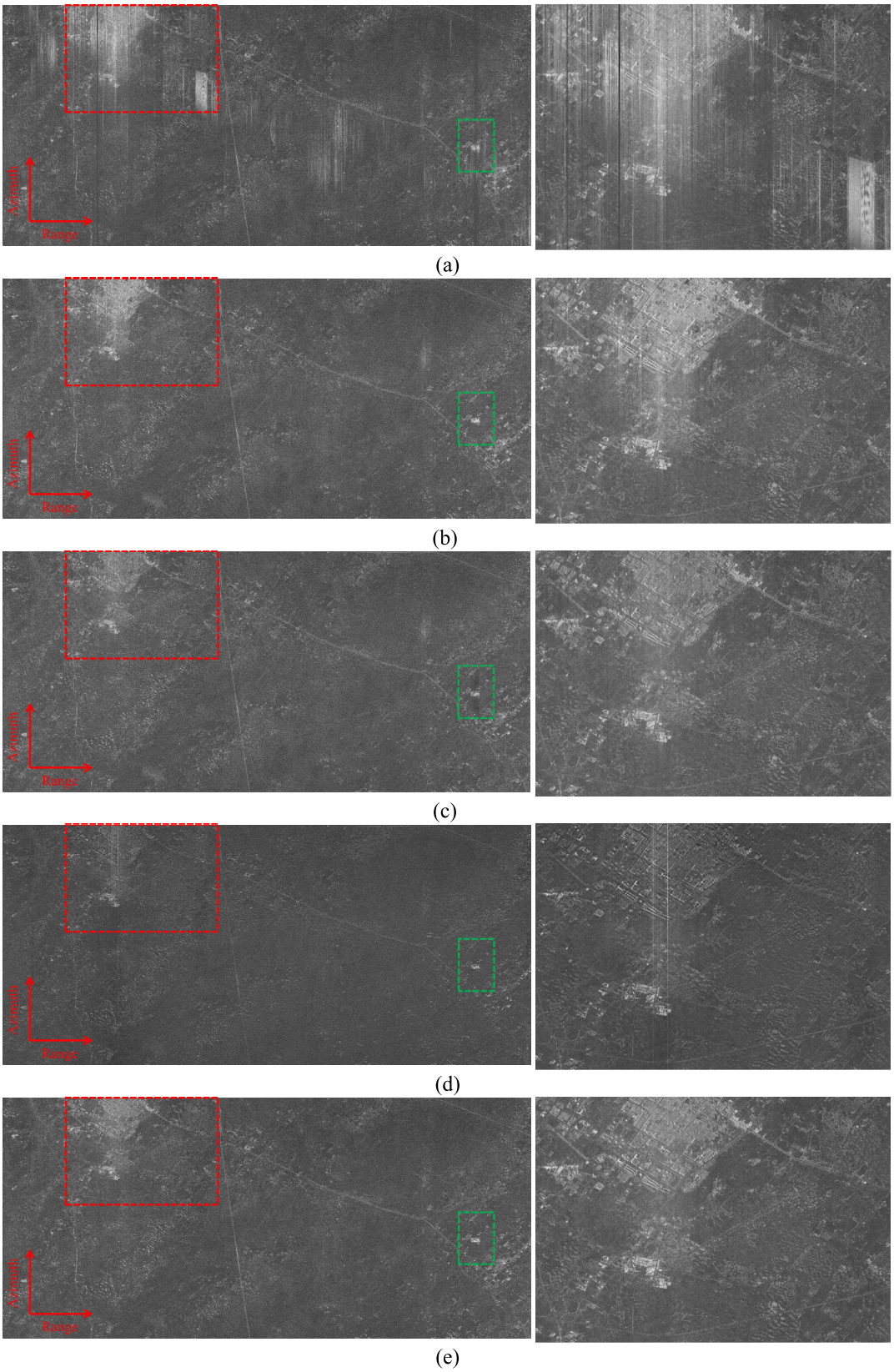


Fig. 13. Experimental result in scenario B. (a) AC [21]. (b) 2D-CFAR [4]. (c) MW [18]. (d) Proposed STFT. (e) Proposed FRFT.

is a 1-D vector of length N_a for FRFT and a 2-D matrix of size $M \times N_a$ for STFT, which results in the latter having slightly higher computational complexity than the former. The whole computational complexity of the proposed STFT and FRFT are respectively

$$T_{\mathrm{STFT}} = O\left(N_r\left(2N_a\log_2 N_a + \mathrm{MN}_a\right)\right) \tag{18}$$

$$T_{\mathrm{FRFT}} = O\left(N_r\left(2N_a\log_2 N_a + \log_2 N_{\mathrm{OUT}} + N_a\right)\right) \tag{19}$$

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where M is the number of frequency bins. In our experiments, the value of M is 128. N_{OUT} is the GESD outlier number for each range cell, which meets $N_{\mathrm{OUT}} \ll N_a$.

The length of N_r is the main factor determining the duration of several methods. Therefore, the time of STFT and FRFT is relatively less in scenario A. It is these two methods that work directly with the ambiguity areas. However, when applied to all the range cells like scenario B, the proposed method does not have a significant time advantage.

V. Conclusion

Serious range ambiguity affects image quality and subsequent applications. Most existing range ambiguity suppression schemes originate from the transceiver system level or post-processing of the original echo data relying on prior information. However, raw data can be challenging for users to obtain, whereas SLC products are typically accessible. To tackle this challenge, this article introduces a method of range ambiguity suppression based on FRFT. It is worth noting that the proposed method is not suitable for the echo domain because of the poor time-frequency concentration property in the raw echo. The superiority of the article is highlighted below.

1) An analytical model in the image domain is constructed for characterizing range ambiguity. With consideration of the model and point target simulation, the characteristic disparities between the desired signal and range ambiguity are summarized.
2) The proposed method aids in determining whether the ghost type is from range ambiguity. The experiments show the proposed method processed on SLC images can achieve similar performance as the traditional judging method on the echo.
3) The proposed method can quickly detect and remove the range ambiguity without prior information. Compared with existing post-processing methods, the performance of RASR and time of our proposed method have achieved satisfactory results without estimating the range ambiguity regions. The proposed method meets the requirements of ground system processing.

Nonetheless, the proposed method requires traversing all range cells of the scenario, which may increase processing time as the scene size increases. If the range ambiguity regions can be obtained in advance, the processing efficiency can be further enhanced. Meanwhile, the current phenomena and conclusions are solely based on strip mode measured data. Therefore, future research will rely on the accurate range ambiguity regions estimation and exploring other imaging modes.

Appendix A

This section is a supplement to the part of Section II about the range ambiguity model from raw echo to SLC images, which helps provide a more detailed explanation of the phenomena.

First, (3) can be written as (A1)

$$s_{\mathrm{ra}}(\tau,\eta) = A_{\mathrm{ra}}\omega_a(\eta-\eta_c)\omega_r\left(\tau-\frac{2R_r(\eta)}{c}\right)\exp\left(-j\frac{4\pi R_{ca}}{\lambda}\right)\exp\left(-j\pi K_{ar}\left(1+\frac{R_{\mathrm{ra}}^2}{2R_{ca}^2}\right)(\eta+\mathrm{PRT})^2\right)\times\exp\left(j\pi K_r\left(\tau-\frac{2R_r(\eta)}{c}\right)^2\right). \tag{A1}$$

In RDA imaging, (A1) can be written as (A2) by range FFT

$$s_{\mathrm{ra}}(f_r,\eta) = A_{\mathrm{ra}}\omega_a(\eta-\eta_c)\omega_r\left(\frac{f_r}{K_r}\right)\exp\left(-j\frac{4\pi R_{ca}}{\lambda}\right)\exp\left(-j\pi K_{ar}\left(1+\frac{R_{\mathrm{ra}}^2}{2R_{ca}^2}\right)(\eta+\mathrm{PRT})^2\right)\exp\left(-j\pi\frac{f_r^2}{K_r}\right)\exp\left(-j4\pi f_r\frac{R_r(\eta)}{c}\right). \tag{A2}$$

Then, (A2) can be written as (A3) through range pulse compression and IFFT

$$s_{\mathrm{ra}}(\tau,\eta) = A_{\mathrm{ra}}A_1\omega_a(\eta-\eta_c)p_r\left(\tau-\frac{2R_r(\eta)}{c}\right)\exp\left(-j\frac{4\pi R_{ca}}{\lambda}\right)\times\exp\left(-j\pi K_{ar}\left(1+\frac{R_{\mathrm{ra}}^2}{2R_{ca}^2}\right)(\eta+\mathrm{PRT})^2\right). \tag{A3}$$

Equation (A3) can be written as (A4) by azimuth FFT

$$s_{\mathrm{ra}}(\tau,f_\eta) = A_{\mathrm{ra}}A_1\omega_a(f_\eta-f_{\eta c})p_r\left(\tau-\frac{2R_{\mathrm{rrd}}(f_\eta)}{c}\right)\exp\left(-j\frac{4\pi R_{ca}}{\lambda}\right)\times\exp\left(-j\pi K_a\left(1+\frac{R_{\mathrm{ra}}^2}{2R_{ca}^2}\right)\left(-\frac{f_\eta}{K_a}+\mathrm{PRT}\right)^2\right) \tag{A4}$$

where $R_{\mathrm{rrd}}(f_\eta)$ is the slant range of the RCM term $D = 1-\lambda^2 f_\eta^2/8V_r^2$.

Based on Taylor expansion, $R_r(\eta)$ and $R_{\mathrm{rrd}}(f_\eta)$ is equal to

$$R_r(\eta) \approx R_{ca}+\frac{V_r^2(\eta+\mathrm{PRT})^2}{2R_{ca}} = R_{ca}+\frac{V_r^2}{2R_{ca}}\eta^2+\frac{V_r^2}{R_{ca}}\eta\mathrm{PRT}+\frac{R_{\mathrm{ra}}^2}{2R_{ca}} \tag{A5}$$

$$R_{\mathrm{rrd}}(f_\eta) \approx R_{ca}+\frac{V_r^2}{2R_{ca}}\frac{f_\eta^2}{K_a^2}-\frac{V_r^2}{R_{ca}}\frac{f_\eta}{K_a}\mathrm{PRT}+\frac{R_{\mathrm{ra}}^2}{2R_{ca}}. \tag{A6}$$

Then, the RCM term will be compensated by RCMC. $p_r(\cdot)$ will be interpolated through $\sin c$. For range ambiguity, the RCM term still exists, and the range ambiguity after RCMC can be written as the following equation:

$$s_{\mathrm{ra}}(\tau,f_\eta) = A_{\mathrm{ra}}A_1\omega_a(f_\eta-f_{\eta c})\times p_r\left(\tau+\frac{2R_{rd}(f_\eta)}{c}-\frac{2R_{\mathrm{rrd}}(f_\eta)}{c}\right)\exp\left(-j\frac{4\pi R_{ca}}{\lambda}\right)\exp\left(-j\pi K_a\left(1+\frac{R_{\mathrm{ra}}^2}{2R_{ca}^2}\right)\times\left(-\frac{f_\eta}{K_a}+\mathrm{PRT}\right)^2\right). \tag{A7}$$

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Then, by azimuth pulse compression, the azimuth envelope becomes $\sin c$ function $p_a(\cdot)$ with a lower sidelobe. Equation (A7) can be written as (A8) respectively,

$$\begin{aligned} & s_{\mathrm{ra}}\left(\tau, f_\eta\right) \cdot h_a\left(\tau, f_\eta\right) \\ & = A_{\mathrm{ra}} A_1 p_r\left(\tau+\frac{2 R_{rd}\left(f_\eta\right)}{c}-\frac{2 R_{\mathrm{rrd}}\left(f_\eta\right)}{c}\right) \\ & \omega_a\left(f_\eta-f_{\eta c}\right) \exp\left(-j \frac{4 \pi R_{ca}}{\lambda}\right) \exp\left(j \pi \frac{f_\eta^2}{K_a}\right) \\ & \exp\left(-j \pi K_a\left(1+\frac{R_{\mathrm{ra}}^2}{2 R_{ca}^2}\right)\left(\left(\frac{f_\eta^2}{K_a^2}+\mathrm{PRT}^2-2 \frac{f_\eta}{K_a} \mathrm{PRT}\right)\right)\right) \end{aligned} \tag{A8}$$

where $h_a(\tau, f_\eta) = \exp(j\pi(f_\eta^2/K_a))$.

Finally, (A8) can be written as (5) respectively through azimuth IFFT.

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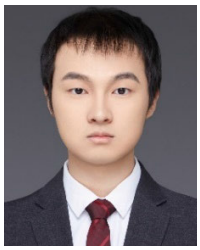
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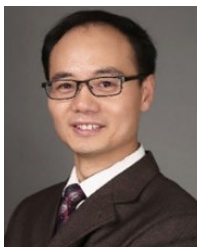
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